

Introduction

Def (Asset): Financial products like stocks, bonds, options

Goal: Find the correct price of assets

Def (Arbitrage): Profiting from price mismatch

Asset Classes

- Bonds (zero coupon): You give the govt. \$X and get paid back $X(1+\frac{R}{n})$, over n years at interest rate R
- Stocks: buying shares gives you fractional ownership of company
- Derivatives: asset based on underlying security (usually stock)
 - Option: contract that gives you the right but not the obligation to buy or sell a stock (at time T , for price K)
 - Future: contract that obliges you to buy or sell the stock at time T for price S , also known as a forward contract

Present and Future Value

If the interest rate is a constant r

$$F(t) = P \left(1 + \frac{r}{n}\right)^{nt}$$

$F(t)$ = future value of \$P (present-value) in t years

n is the # of compounding periods in a year ($n=12$, monthly)

For continuous compounding: $F(t) = Pe^{rt}$

Mortgages and Annuities

- Mortgage: You borrow \$L, pay it back in N equal monthly payments of P
- We work in time units of months, r is annual interest rate

$$L = \sum_{k=1}^N P \left(1 + \frac{r}{12}\right)^{-k}, \quad B_L = \left(L - P \sum_{k=1}^N \left(1 + \frac{r}{12}\right)^{-k} \right) \left(1 + \frac{r}{12}\right)^N$$

- Annuity: a retirement plan where you deposit an amount P every year into an account so that, after retirement you can withdraw W_L dollars in L years for each year $E = 1, \dots, N$ set by maturity
- Let D = PV of amount at retirement, M = maths from retirement

$$D = \sum_{i=1}^N W_i (1+r)^{-i} = \sum_{i=1}^N P \left(1 + \frac{r}{12}\right)^i$$

Variable Interest Rates

Def (Spot rate): The interest rate to loan made at $t=0$, repaid by $t=T$, denoted $r(t,T) \approx$ interest rate of zero-coupon bonds

Def (Forward interest rate): The no arbitrage interest rate of a loan made at $t=T_1$, paid back by $t=T_2$, denoted $r(T_1, T_2)$

$$r(T_1, T_2) = \frac{r(T_1, T_2)T_2 - r(T_1, T_1)}{T_2 - T_1} \quad \& \quad r(T) := \lim_{T_2 \rightarrow T} r(T_1, T_2) = \frac{d}{dT} (r(T_1, T))$$

$$r(T_1, T_2) = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} r_t dt \Rightarrow \text{time avg. of instant rates}$$

Bonds

Def (Zero Coupon Bond): A promise to pay \$P now in exchange for \$F after N years: $F(1+r(t, N))^N = P$, F = face value bond

Def (Coupon): You get fraction cF every year $P = P_0 + cF \sum_{i=1}^N (1+r(t, N))^{-i}$

Def (Yield of Coupon Bond): R s.t. $P = F(1+R)^{-N} + cF \sum_{i=1}^N (1+R)^{-i}$

Simple Model of Stocks

At each time t , the stock goes up w.p. p_u or down w.p. $p_d = 1-p_u$
 Stock price $S_{t+1} = 2S_t$ where $Z = \begin{cases} u & \text{w.p. } p_u \\ d & \text{w.p. } p_d \end{cases}$ and $u, d > 1$

Portfolio: Tuple $h = (x, y)$, x bonds, y stocks, with value $V_t = x(1+r(t, T))^t y S_t$
 Here x, y can be fractional or negative and represent starting values

Def (Short Selling): Borrow shares, then immediately sell them, invest the proceeds, in this case $q < 0$

Prop. (Arbitrage Condition): $V_t^0 = 0$, $V_t^0 \geq 0$ w.p. 1, $P(V_{t+1}^0 > 0) > 0$ for $t \geq 1$

Thm. There are no arbitrage opportunities $\Leftrightarrow d \leq 1+R \leq u \Leftrightarrow \exists q_u, q_d \in [0, 1]$
 $\Leftrightarrow \exists q_u, q_d \in [0, 1]$ s.t. $q_u q_d = 1$ and $1+R = q_u u + q_d d$

Under no arbitrage we have $q_u = \frac{1+R-d}{u-d}$ & $q_d = \frac{u-(1+R)}{u-d}$

Def (Risk-Neutral Measure Q): $Q(z=1) = q_u$, $Q(z=d) = q_d$ s.t.

$$E_Q(z) = q_u u + q_d d = 1+R \quad \& \quad E_Q\left[\frac{V_{t+1}}{S_{t+1}} \middle| \mathcal{F}_t\right] = X(1+R)^{-t} y S_t = E_Q(z) = (1+R) V_t^0$$

Proposition (Martingale property): $E_Q[V_{t+1} | \mathcal{F}_t] = q_u u + q_d d$

$$\Rightarrow E_Q[S_{t+1} | \mathcal{F}_t] = (1+R) S_t$$

Binomial Option Pricing

Option parameters: Price p of contract, expiring date T , strike price K
 Call option: contract provides right to buy the underlying stock at time $t=T$ for price K
 Put option: right to sell ...

The above describes European options, for American options we can exercise (buy/sell underlying) at any time $t \leq T$.

Rest of this section concerns European options

$$\text{Payoff of option: } \phi(S_T) = \max(S_T - K, 0) \text{ or } \max(K - S_T, 0)$$

Def (Completeness of Market): For any contingent claim $\phi(S_T)$ there's a self-financing portfolio h s.t. $V_t^h = \phi(S_t)$

Def (Self-financing portfolio): $X(t) + y(t)S_t = X(t+1) + y(t+1)S_{t+1}$
 No outside cash injections, change in value only from stock price changes and bond returns due to interest

Prop (One-period model): Suppose $T=1$. The market is complete

then we can find the portfolio $h=(x, y)$ s.t. $V_t^h = \phi(S_t)$ if $S_0 = S$ by solving $x(1+r) + yS = \phi(S_u)$ & $x(1+r) + yS = \phi(S_d)$

$$\Rightarrow \begin{cases} x = \frac{1}{1+r} \frac{\phi(S_u) - \phi(S_d)}{u-d} \\ y = \frac{\phi(S_u) - \phi(S_d)}{S(u-d)} \end{cases}$$

$$\text{No arbitrage} \Rightarrow V_0^h = \frac{1}{1+r} E_Q[V_1^h | S_0=S] = \frac{1}{1+r} E_Q[\phi(S_1)]$$

V_0^h = option price since options can be replicated by stock + bond portfolio at payout time.

Denote by $V(S, t)$ = the price of a call at $t=0$ with underlying price S
 $V(S, t) = \frac{1}{1+r} (q_u \phi(S_u) + q_d \phi(S_d))$

Prop (Multi-period model): Option expiring date $T > 1$. In this

we define arbitrage in the option market as $V_0^h = 0$ but $V_T^h \geq 0$ w.p. 1 and $V_T^h > 0$ w.p. $p > 0$. Completeness holds: Suppose $S_{T-1} = S_{t+1} d^{T-1-t}$ (stock followed a particular up/down seq.) Then for $V_t^h = \phi(S_t)$ we have $S_T = 2S_{T-1}$ so:

$$X(1+R)^T + y = \phi(S_{T-1}) \quad \& \quad X(1+R)^T + y d S_{T-1} = \phi(S_{T-1} d)$$

$$\begin{cases} x = \frac{1}{(1+R)^T} \left[\frac{u\phi(S_{T-1}u) - d\phi(S_{T-1}d)}{u-d} \right] \\ y = \frac{\phi(S_{T-1}u) - \phi(S_{T-1}d)}{S_{T-1}(u-d)} \end{cases}$$

Note: The stocks movements are independent each day so:

$$Q(\text{stock goes up } k \text{ times by time } T) = \binom{T}{k} q_u^k q_d^{T-k} \quad (\text{Binomial dist!})$$

$$\begin{aligned} \text{The option price } V(0, S) &= V_0^h = \frac{1}{1+R} E_Q[V_1^h | S_0] \\ &= \frac{1}{(1+R)^T} E_Q[E_Q[V_T^h | \mathcal{F}_t] | S_0] \\ &= \frac{1}{(1+R)^T} E_Q[V_T^h | S_0] \\ &= \frac{1}{(1+R)^T} E_Q[\phi(S_T) | S_0] \end{aligned}$$

Notation: $V_k(t)$ = value of portfolio at time t if the stock goes up k times by time t

$$V_T(k) = \phi(S_t u^k d^{T-k}) \quad \text{using this at the leaf of tree, we can recurse back to find the positions}$$

Goal: Find $X_k(t), Y_k(t)$. Consider time t , if stock went up/down

$$(1+R)X_k(t) + Y_k(t)u S_t = V_{k+1}(t+1)$$

$$(1+R)X_k(t) + Y_k(t)d S_t = V_{k-1}(t+1)$$

$$X_k(t) = \frac{1}{1+R} \frac{V_{k+1}(t+1) - V_{k-1}(t+1)}{u-d} \quad \& \quad Y_k(t) = \frac{V_{k+1}(t+1) - V_{k-1}(t+1)}{S(u-d)}$$

Thm (Option price, $T > 1$): We have an option with expiring time T , payoff $\phi(S_T)$, then the price of the option now if $S_t = S$:

$$\begin{aligned} V(0, S) &= \frac{1}{(1+R)^T} \sum_{k=0}^T \phi(S u^k d^{T-k}) Q(\text{stock up } k \text{ times by } T) \\ &= \frac{1}{(1+R)^T} \sum_{k=0}^T \binom{T}{k} q_u^k q_d^{T-k} q_u^k q_d^{T-k} \end{aligned}$$

Ex: We can find prices at intermediate times t conditional on S_t for $t < T$

$$\begin{aligned} V(t, S_t) &= \frac{1}{(1+R)^{T-t}} E_Q[\phi(S_T) | S_t] \\ &= \frac{1}{(1+R)^{T-t}} \sum_{k=0}^{T-t} \binom{T-t}{k} \phi(S_t u^k d^{T-t-k}) q_u^k q_d^{T-t-k} \end{aligned}$$

Call $\phi(S_t u^k d^{T-t-k}) = \max\{S_t u^k d^{T-t-k} - K, 0\} \Rightarrow$ find min k s.t. $S_t u^k d^{T-t-k} > K$
 strike price $\Rightarrow$ as k inc. the term gets bigger $\Rightarrow$

Investing

Efficient Market Hypothesis: all public information is reflected in current stock prices and $\Rightarrow$ this one and cannot consistently make abnormal profits

ϕ_t = all public information available at time t
 $\epsilon_{t+1} = S_{t+1} - E[S_{t+1} | \phi_t]$ (magnitude of mispricing)

Mathematically: Efficient market $\Rightarrow E[\epsilon_{t+1} | \phi_t] = 0$

- Rational investors balance return vs risk: high risk, high reward. Given same level of risk, they favor higher reward, and given the same return/reward, they favor lower risk
- Risk aversion is described by a utility function

Def (Utility function): A function that describes how much utility we assign personally to a given wealth level u , denoted $u(u)$

- Someone who is initially averse to risk has $u(u) = u$
- Someone who is equally upset at losing 1% of their wealth regardless of the dollar amt. has $u(u) = \log u$
- $u(u)$ describes your risk aversion; a risk averse person has a utility $u(u) = u$ s.t. $u(u) \geq E[u(u)]$ for any bet u s.t. $E(u) > 0$, which means that they prefer to have u in his account over taking a bet of u
- Risk premium: π s.t. $u(u - \pi) = E(u(u))$
 How much you would have to take to lose bet u
- If $u(u) = \log u$, we compute π (the fraction of u he would rather lose than play a game/bet u)
 $\Rightarrow \log(u - \pi) = E[\log(u + X)]$
 would rather lose "losing fraction π of u fraction π of u than

$$\log(u - \pi) = E[\log(u + X)] \Rightarrow \text{using to bet on stocks, regardless Kelly Criterion: Investor wants to fraction X in stocks, interest rate r . You get $X \times X$, where $X = u$ w.p. p , $X = d$ w.p. $1-p$. Wealth in next period: $W_t = u(1+r) + X \times X$
 $W_t = \prod_{i=1}^t u(1+r + X_i) \Rightarrow \log W_t = \frac{1}{n} (\log u + \sum_{i=1}^n \log(1+r + X_i))$
 $\lim_{n \rightarrow \infty} \log W_t = E[\log(1+r + X)]$ w.p. 1 (want to maximize)
 $\Rightarrow X = \frac{d - r}{u - r}$ (Taylor expand + derivative)$$

CAPM

- Return of stock i : $r_i \sim N(\mu_i, \sigma_i^2)$ ($\text{Var}(r_i) = \sigma_i^2$, $E(r_i) = \mu_i$)
- Portfolio consisting of N stocks $1, \dots, N$ with weights W_i of each stock s.t. $\sum_{i=1}^N W_i = 1$ has mean return $\mu_p = E[\sum_{i=1}^N W_i r_i] = \sum_{i=1}^N W_i \mu_i$
- and variance: $\sigma_p^2 = \text{Var}(\sum_{i=1}^N W_i r_i) = \sum_{i=1}^N W_i^2 \text{Var}(r_i) + 2 \sum_{i < j} W_i W_j \text{Cov}(r_i, r_j)$

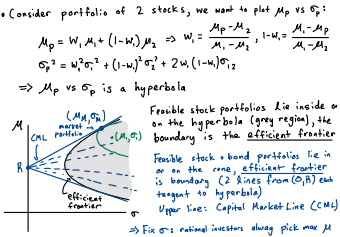
$$\sigma_i^2 := \text{Cov}(r_i, r_i) = E[(r_i - \mu_i)(r_i - \mu_i)] \quad \& \quad \mu_i = \mu_p - \mu_p \sigma_i$$

• Consider portfolio of 2 stocks, we want to plot μ_p vs σ_p^2 :

$$\mu_p = W_1 \mu_1 + (1-W_1) \mu_2 \Rightarrow W_1 = \frac{\mu_p - \mu_2}{\mu_1 - \mu_2}, \quad 1-W_1 = \frac{\mu_1 - \mu_p}{\mu_1 - \mu_2}$$

$$\sigma_p^2 = W_1^2 \sigma_1^2 + (1-W_1)^2 \sigma_2^2 + 2W_1(1-W_1)\sigma_{12}$$

$\Rightarrow \mu_p$ vs σ_p is a hyperbola



$$\mu_{CML} = \frac{\mu_M - R}{\sigma_M} = \max_{W_1} \frac{\mu_p - R}{\sigma_p} = \max_{W_1} \frac{\mu_1 - R}{\sigma_1} = \frac{\mu_1 - R}{\sigma_1}$$

Equation of CML: Goal: Write μ_p in terms of μ . Consider set of feasible portfolios that consist of M and portfolio i

$$\mu = R + \sigma_i \sigma_M \Rightarrow \mu - R = \sigma_i \sigma_M \Rightarrow \mu - R = \sigma_i \sigma_M$$

$$\Rightarrow L \leq \frac{y(u) - \mu}{\sigma} = \frac{y(u) - \mu}{\sigma} \Rightarrow y(u) = \mu$$

$$X^*(u) = \frac{1}{2} \sigma^2 + (1 - \frac{1}{2} \sigma^2) \mu + 2(1 - \frac{1}{2} \sigma^2) \sigma_M \Rightarrow X^*(u) = \mu$$

$$2 \times \frac{d}{d\mu} = 2 \times \frac{d}{d\mu} - 2(1 - \frac{1}{2} \sigma^2) \sigma_M + 2 \sigma_M (1 - \frac{1}{2} \sigma^2)$$

$$\mu_{CML} = \frac{d\mu}{d\sigma} \bigg|_{\sigma=\sigma_M} = \frac{\mu_M - R}{\sigma_M} = \frac{\mu_p - R}{\sigma_p}$$

$$\mu_p = R + \frac{\sigma_p}{\sigma_M} (\mu_M - R) \quad \text{Portfolio } p \text{ allowed to have stocks + bonds}$$

